Math 2210 QUIZ 1

Ex 1 Find the angle between (1,2,0) & (3,-1,1)

$\vec{u}\vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$ $\qquad \cos\theta = \frac{3-2}{\sqrt{5}\sqrt{11}} = \frac{1}{\sqrt{55}}$ $\qquad \theta = \cos^{-1}\frac{1}{\sqrt{55}}$

Ex 2 Find the eqn of the plane through (3,-1,1) with normal direction (1,-4,2)

$x-3-4(y+1)+2(z-1)=0$

$x-4y+2z=9$

Ex 3 Find the arc length of $x=\cos t$ $y=\sin t$ $z=-t$ for $0 \le t \le 2\pi$.

$x' = -\sin t \qquad y' = \cos t \qquad z' = -1$

$$L = \int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t + 1^2}\, dt = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}$$